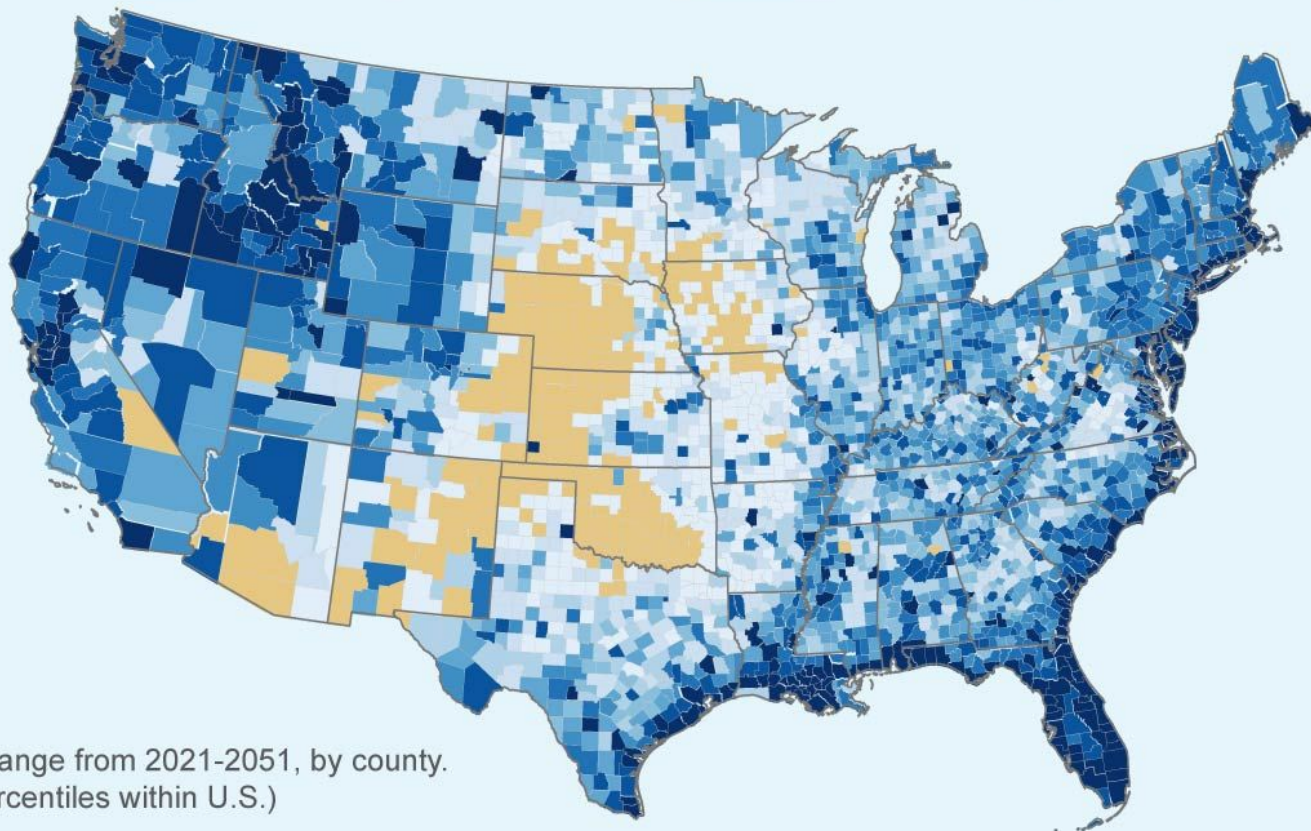


A white sports car is partially submerged in floodwater. Smoke is rising from the engine area. The background shows trees and a street sign.

Exploring OpenFEMA Dataset features most relevant to policy cost

Change in flood risk over the next 30 years

Decreasing risk Increasing risk



Overall risk change from 2021-2051, by county.
(Change in percentiles within U.S.)

parquet

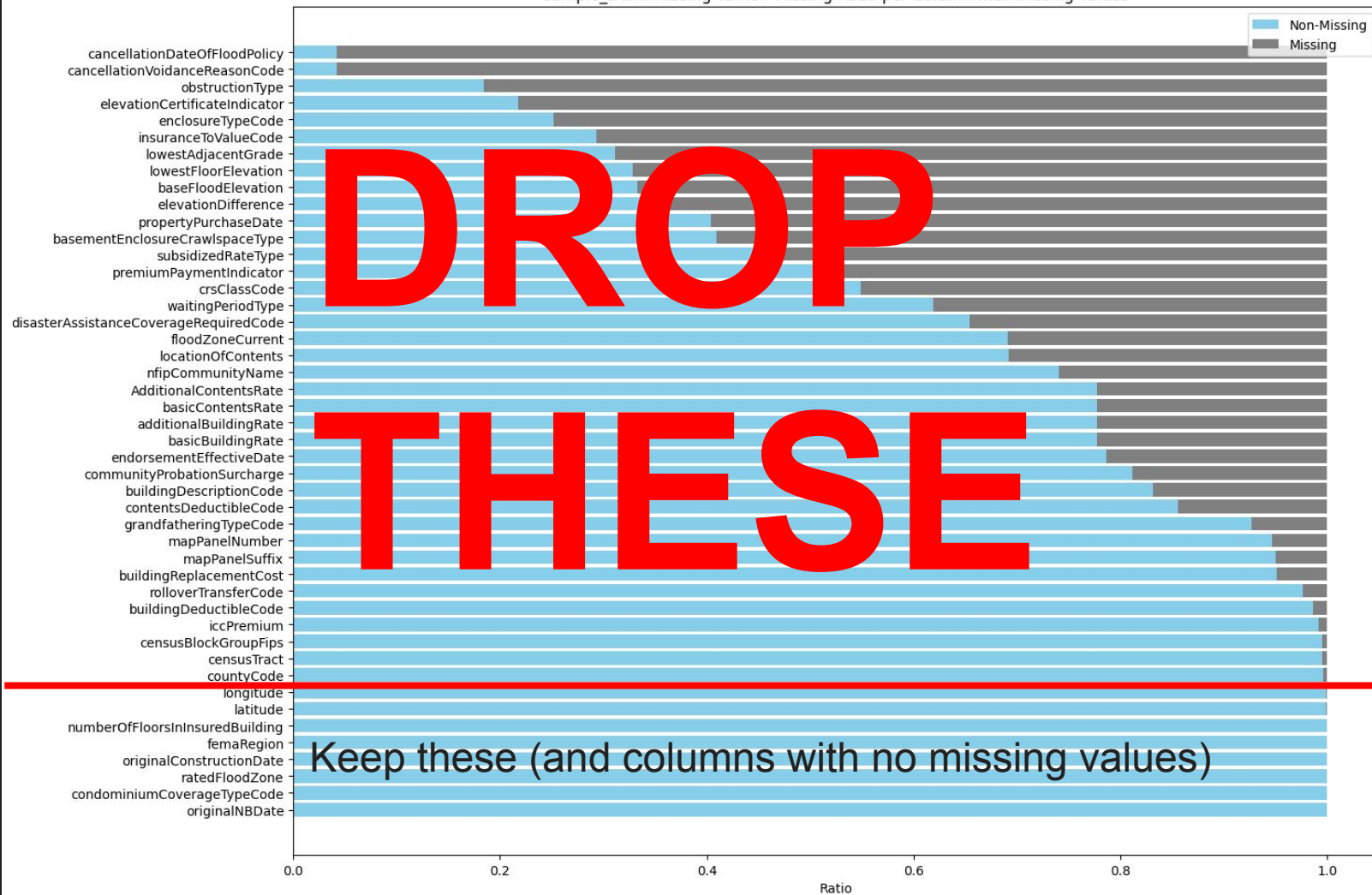
[Link to parquet](#)

72583994

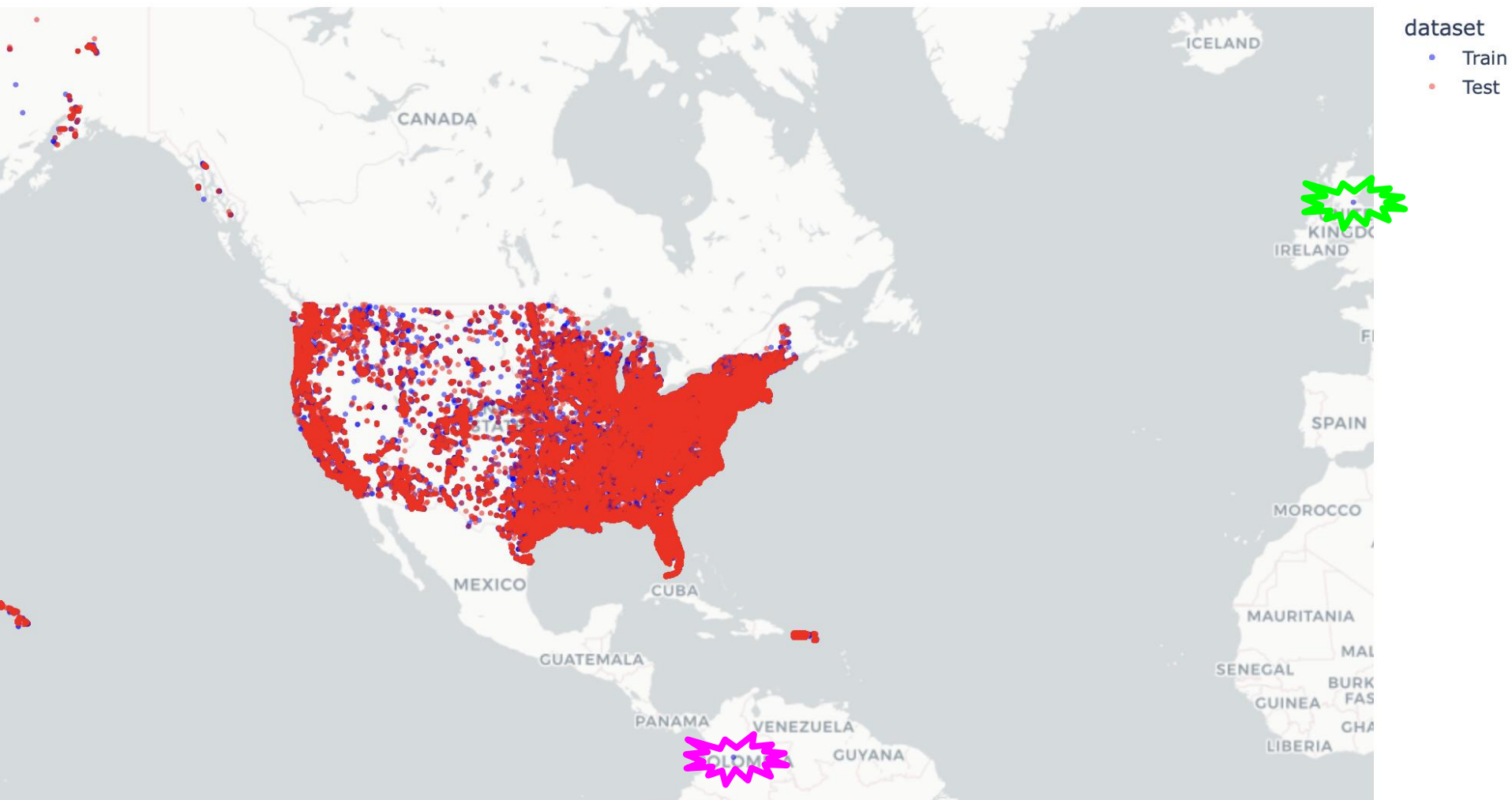
large (500MB - 10GB)

- Over **70 million** records!
- I sampled via hashing
 - ~350,000 for training
 - ~350,000 for testing

sample_train: Missing vs Non-Missing Ratio per Column with missing values



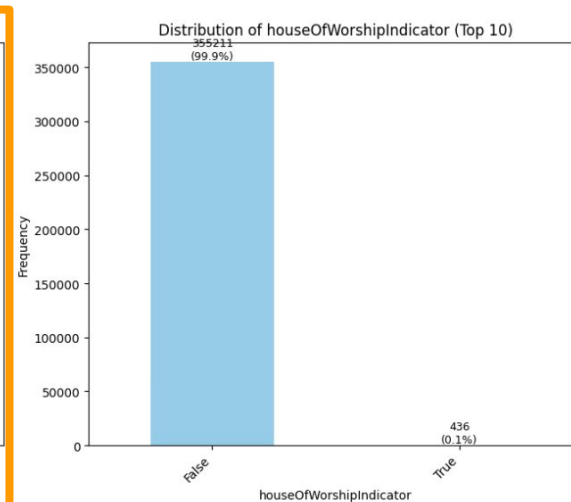
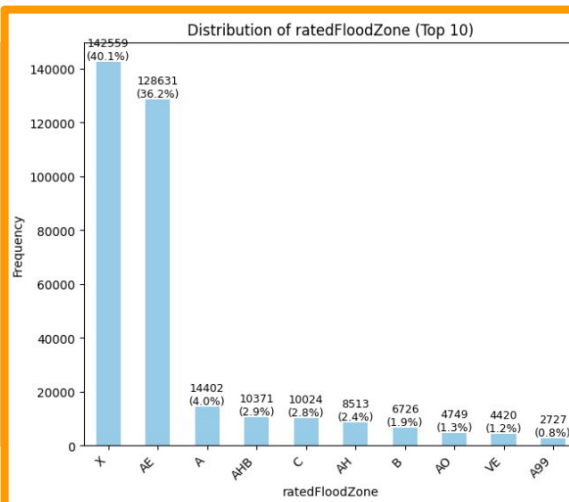
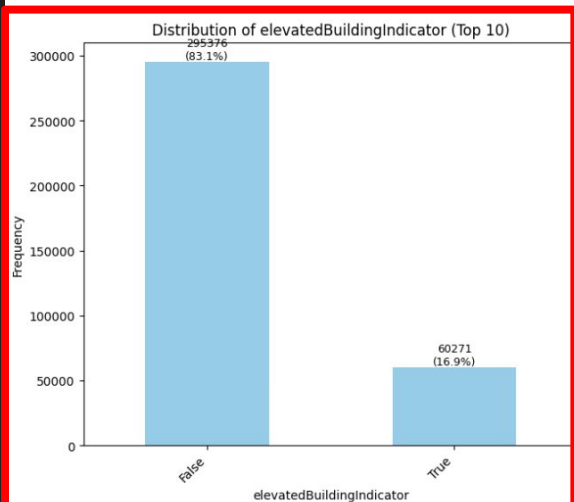
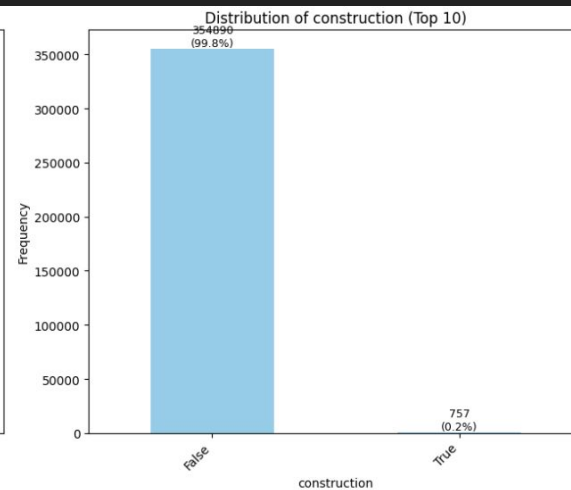
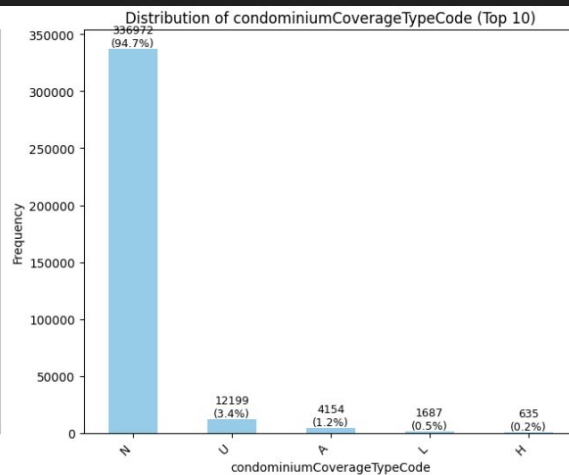
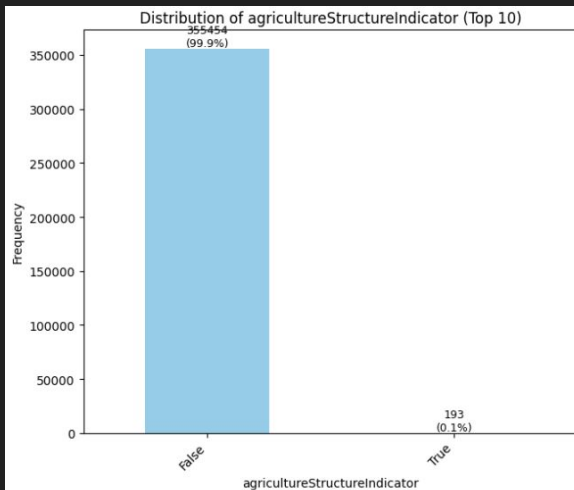
FEMA Geographic Outliers (Blue=Train, Red=Test)

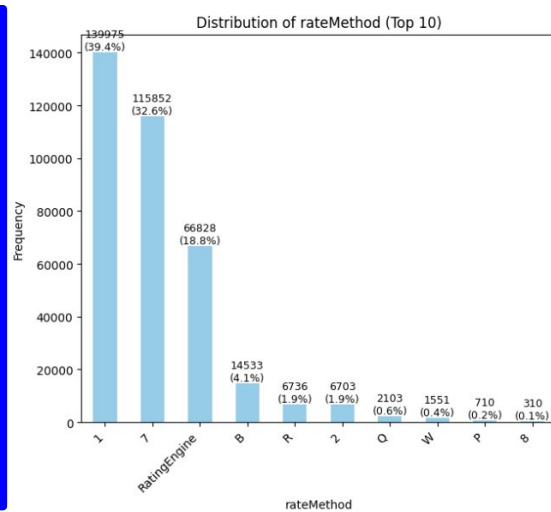
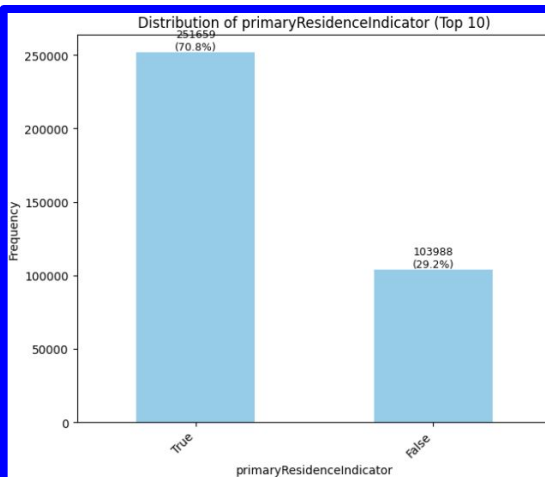
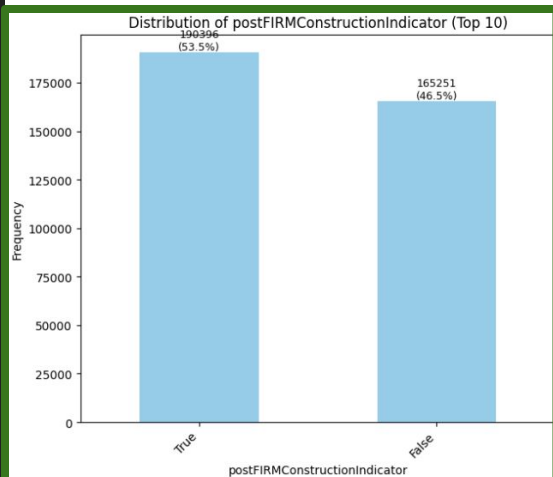
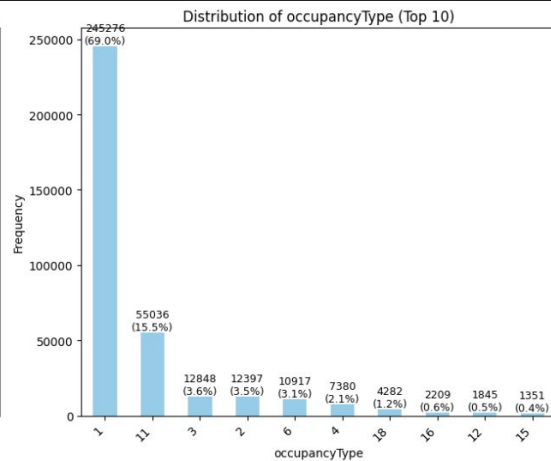
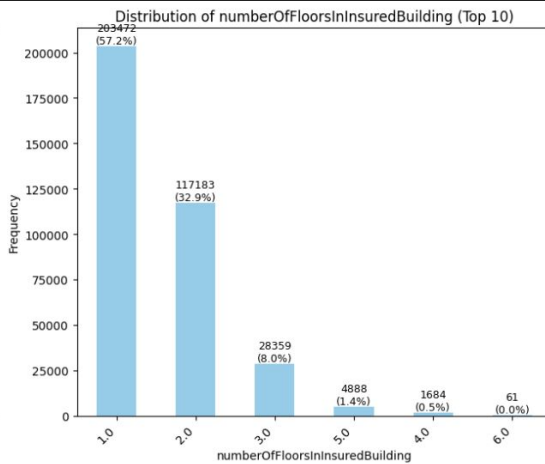
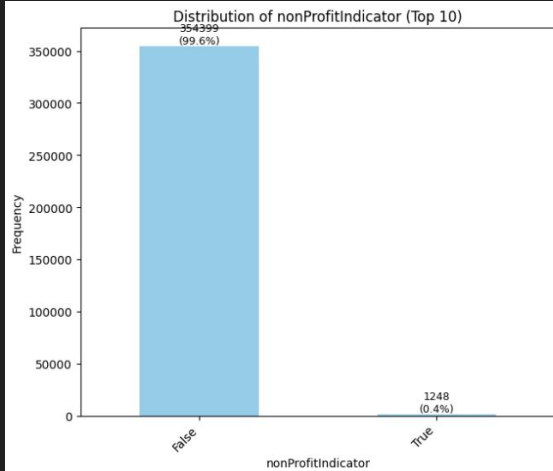


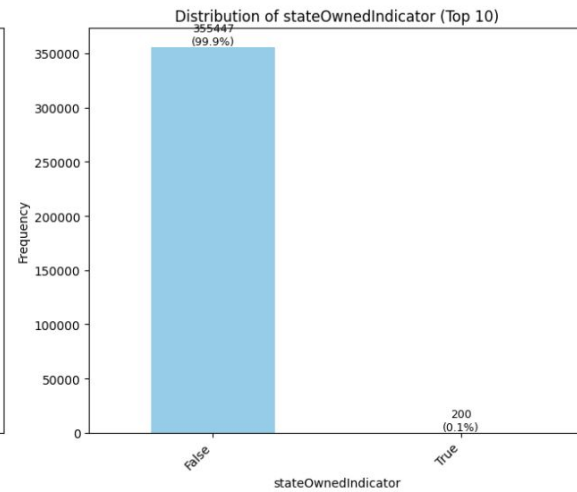
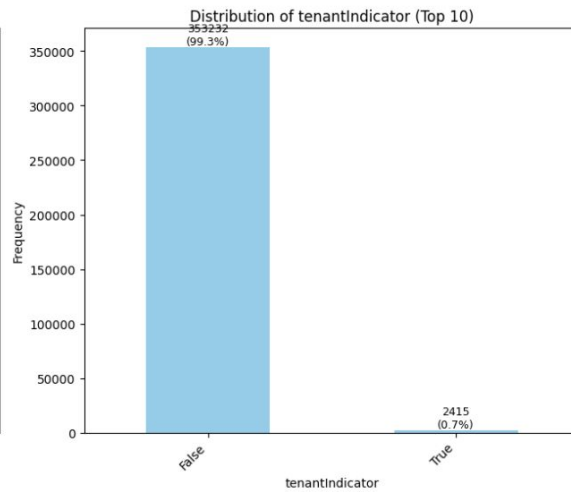
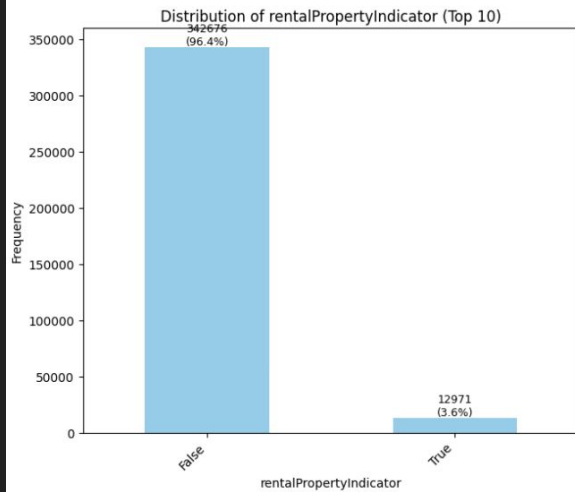
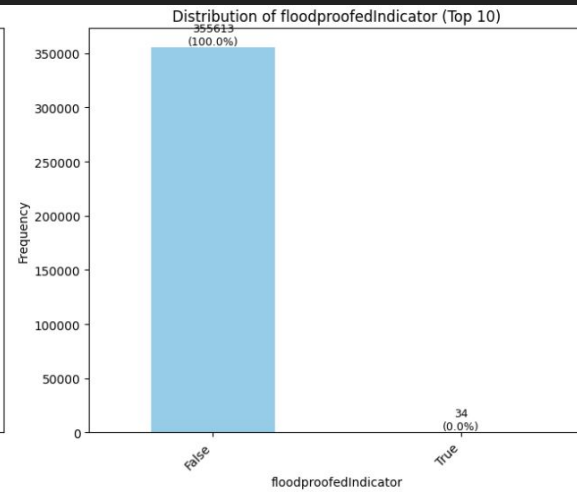
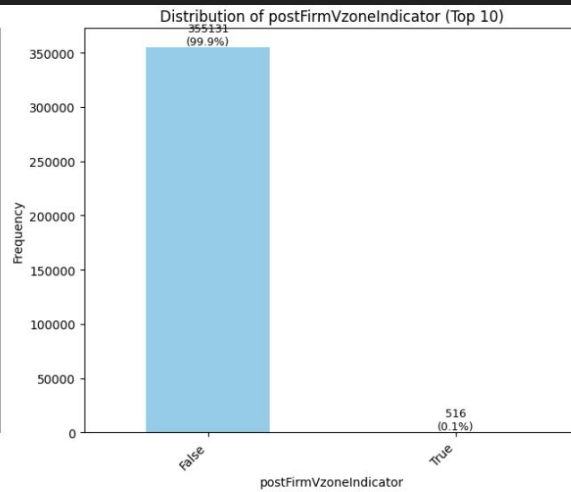
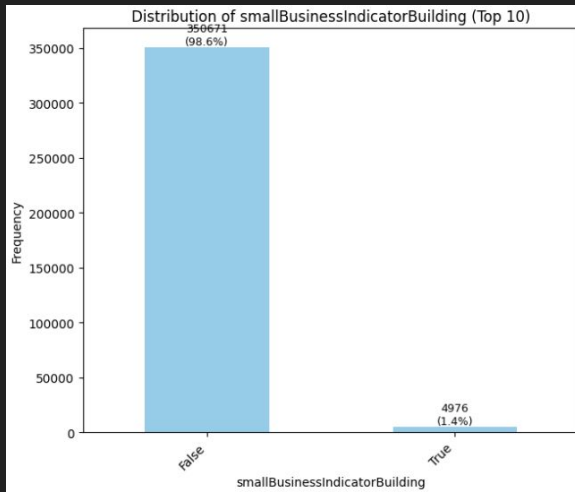
Found 2 rows at these coordinates:

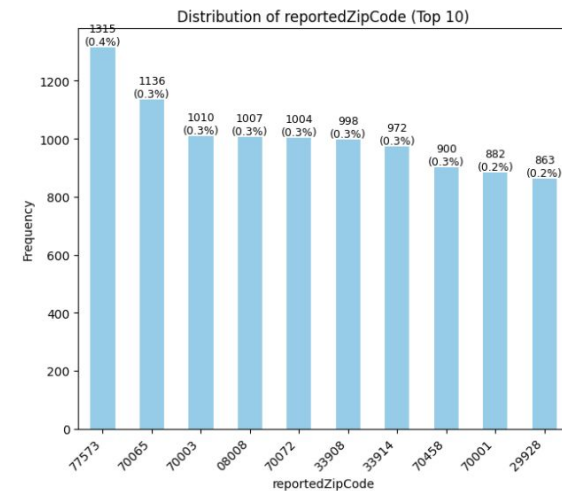
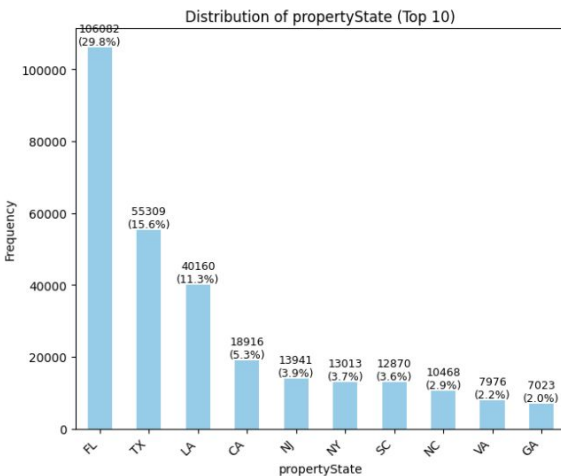
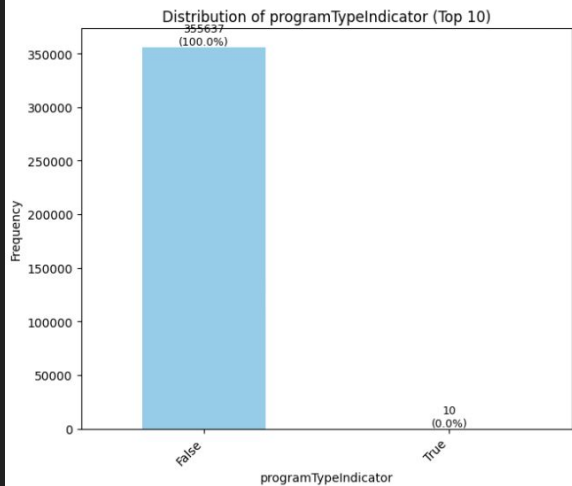
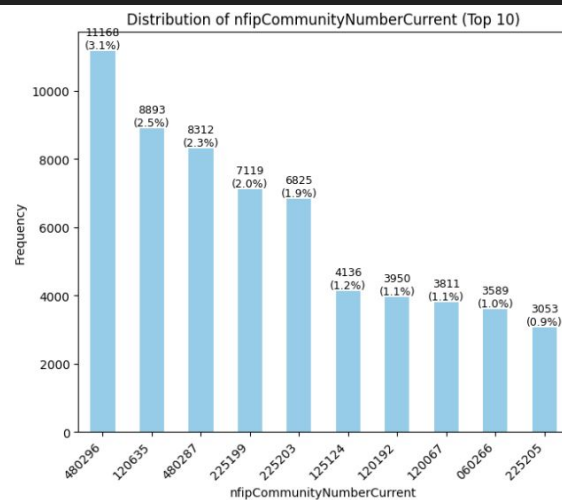
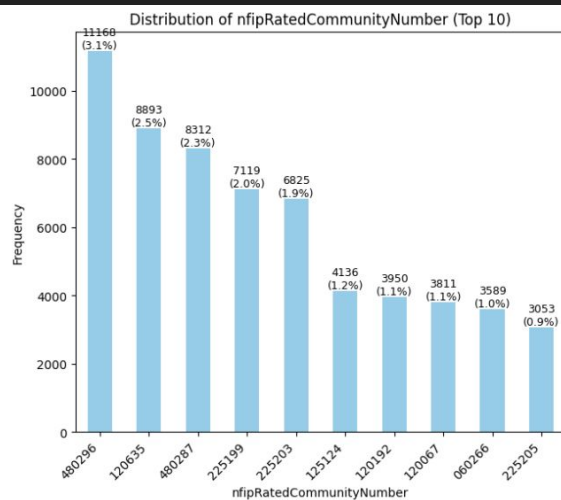
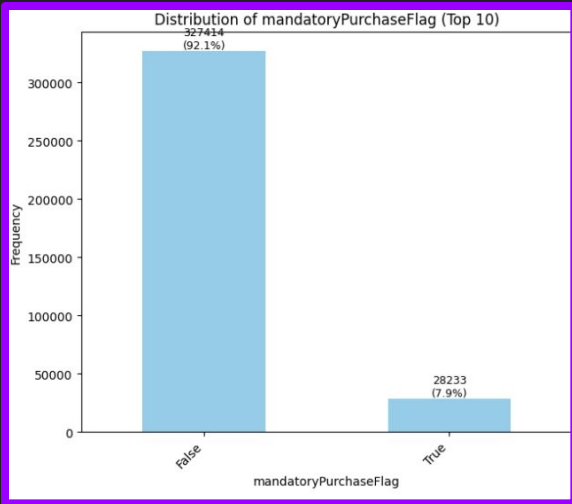
		propertyState	reportedZipCode	latitude	longitude
193990		MS	39759	56.1	-3.2
213838		PR	00949	4.6	-74.1

Dropped along with all
rows missing values



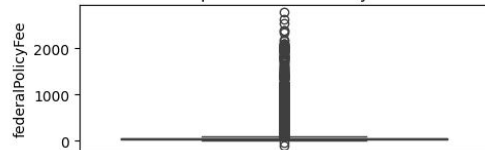




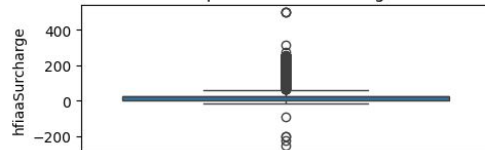


- Many features extremely unbalanced (over 90% the same value)
- To avoid curse of dimensionality, I only chose the most balanced
 - `elevatedBuildingIndicator`
 - `ratedFloodZone` (which I got dummies as X, AE)
 - `postFIRMConstructionIndicator`
 - `primaryResidenceIndicator`
 - `mandatoryPurchaseFlag`

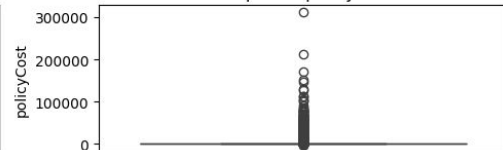
Boxplot of federalPolicyFee



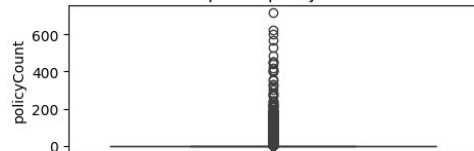
Boxplot of hfiaaSurcharge



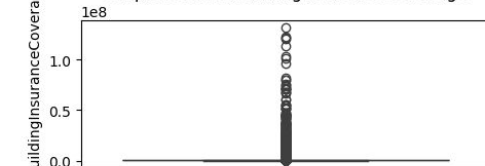
Boxplot of policyCost



Boxplot of policyCount

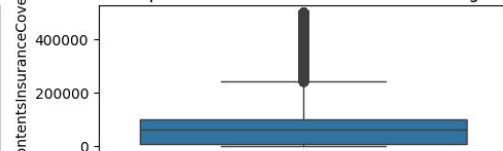


Boxplot of totalBuildingInsuranceCoverage

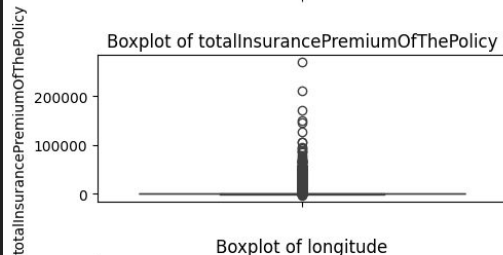


totalContentsInsuranceCoverage

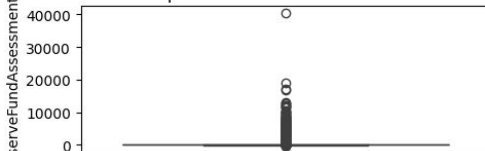
Boxplot of totalContentsInsuranceCoverage



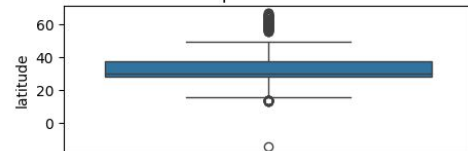
Boxplot of totalInsurancePremiumOfThePolicy



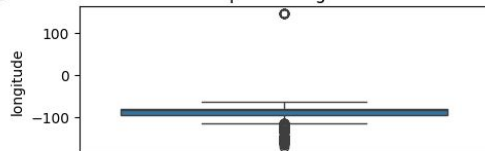
Boxplot of reserveFundAssessment



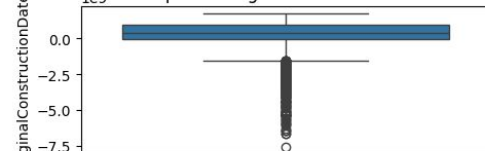
Boxplot of latitude



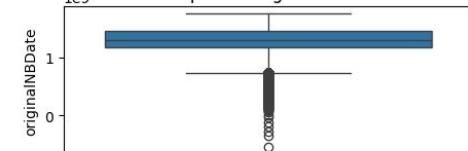
Boxplot of longitude



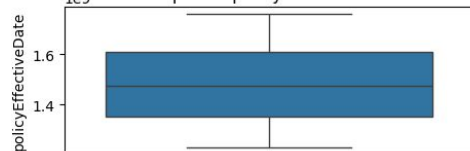
Boxplot of originalConstructionDate



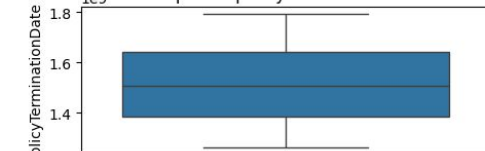
Boxplot of originalINBDate



Boxplot of policyEffectiveDate



Boxplot of policyTerminationDate



Problems

1. Latitude, Longitude
2. Skew
3. Scale

Solution pipeline

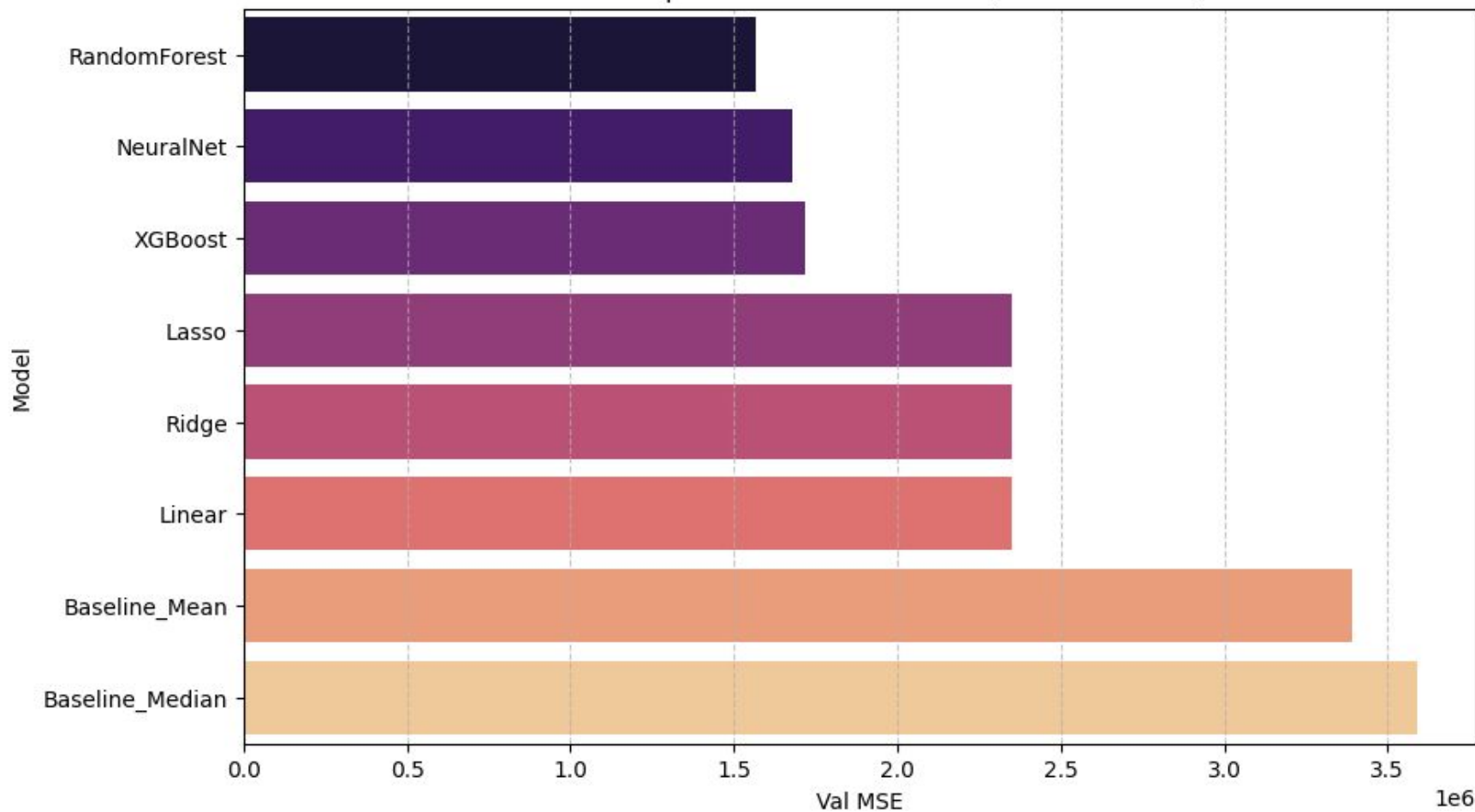
1. Cyclical encoding (3D Cartesian coords)
2. Reflect if left skew, log transform
3. Standard scaler

policyCost	Policy Cost	integer	Calculated in dollars by adding together calculated premium, reserve fund assessment, federal policy fee, and HFIAA surcharge
------------	-------------	---------	--



Multicollinearity danger!

Model Comparison: Validation MSE (Lower is Better)



Winning Model: RandomForest

Test MSE

1779859.7177

Test RMSE

1334.1138

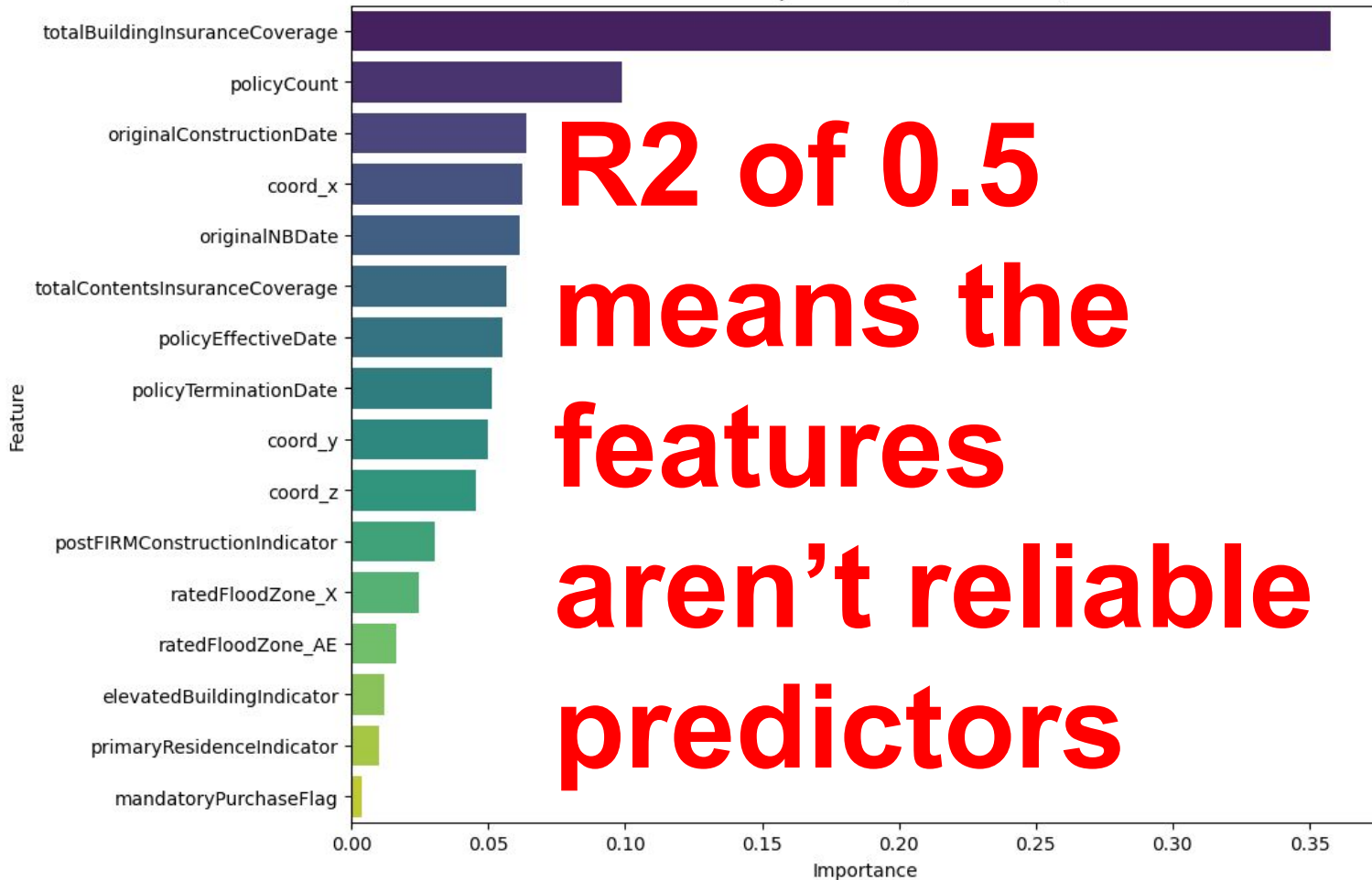
Test MAE

409.1102

Test R2

0.5058

Feature Importances (RandomForest)

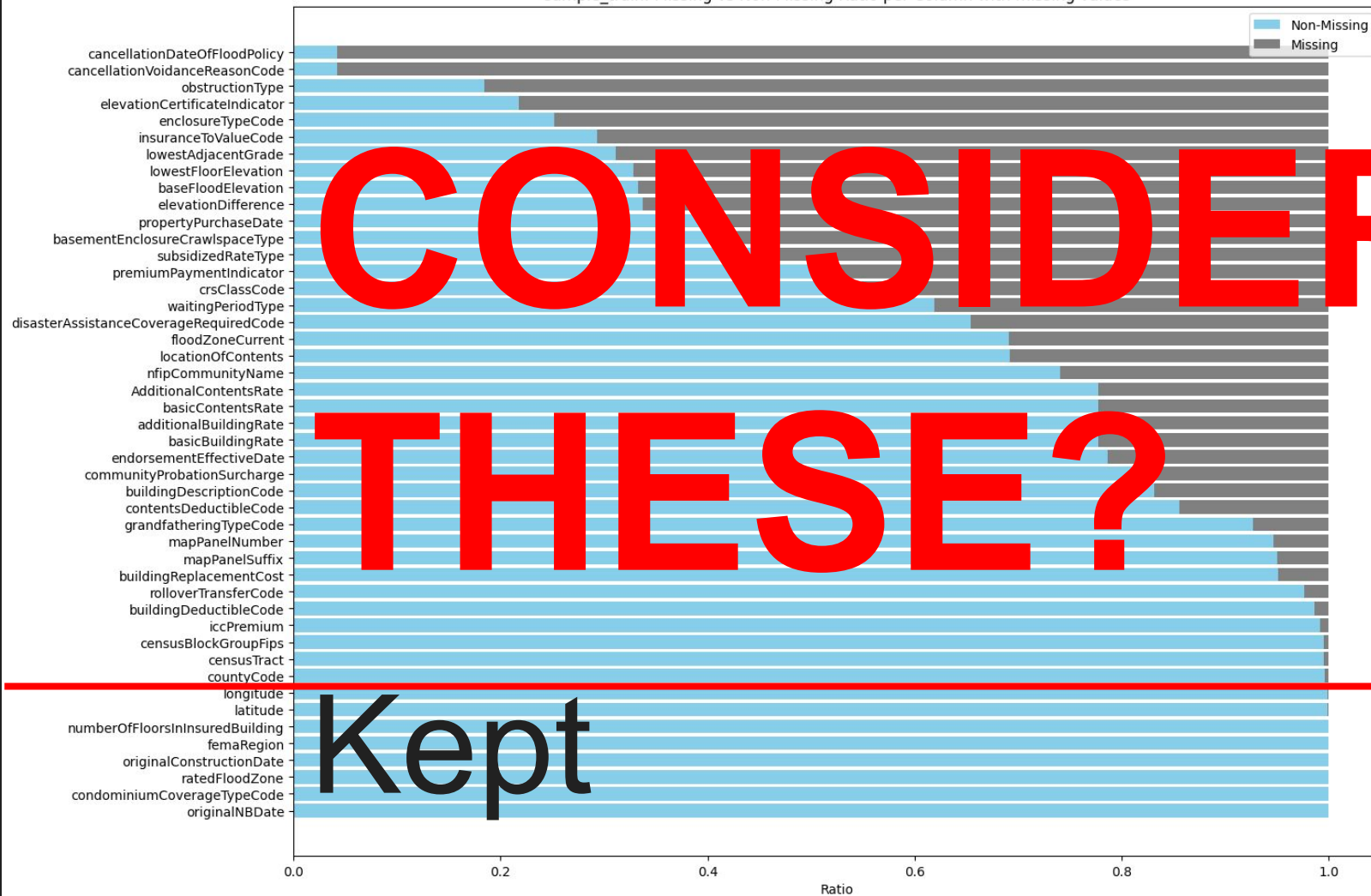


policyCost	Policy Cost	integer	Calculated in dollars by adding together calculated premium, reserve fund assessment, federal policy fee, and HFIAA surcharge
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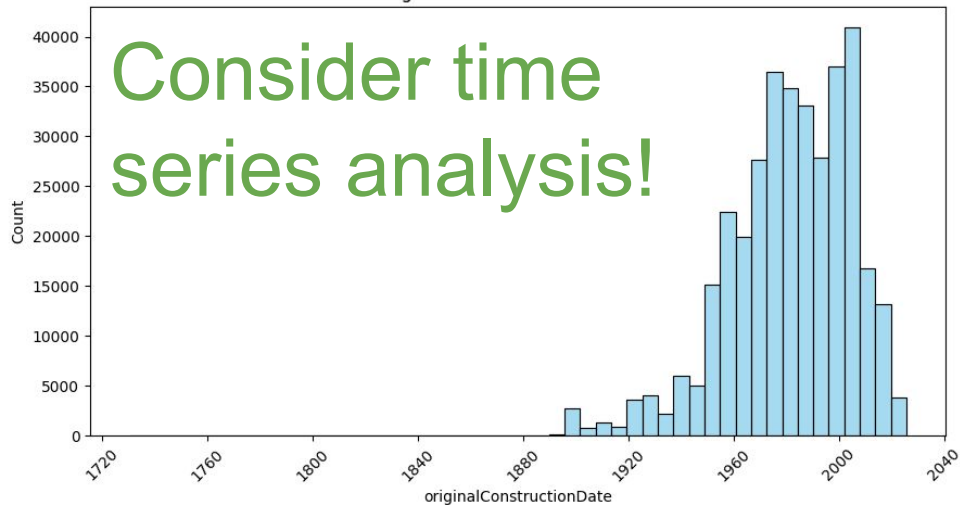


Calculate (don't predict)

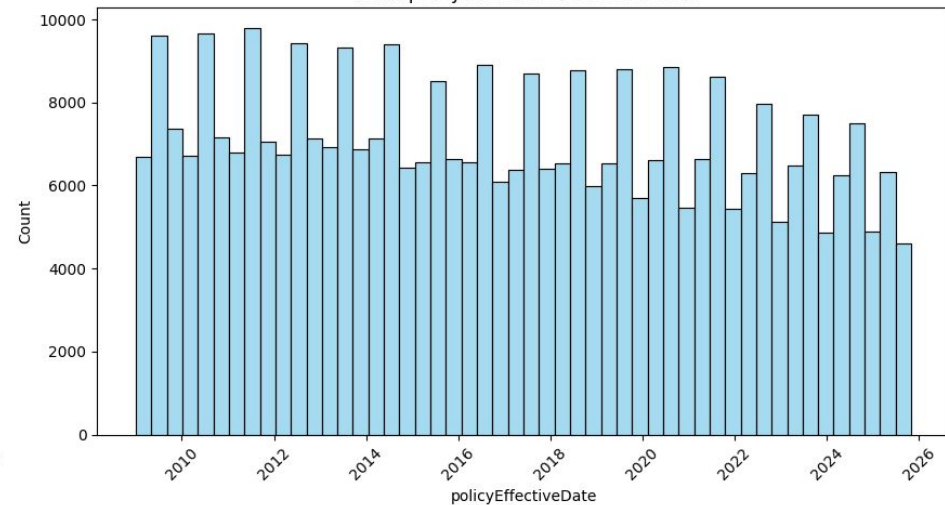
sample_train: Missing vs Non-Missing Ratio per Column with missing values



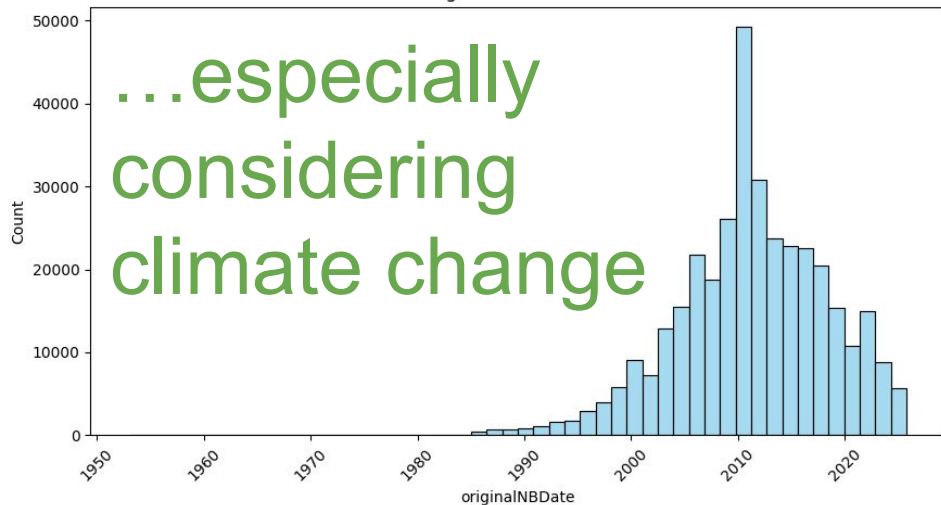
Train: originalConstructionDate Distribution



Train: policyEffectiveDate Distribution



Train: originalNBDate Distribution



Train: policyTerminationDate Distribution

